

IT Research Methodology

Introduction to Research Methodology

Image Source: Research Methodology

Prepared by: CHEN SOVANN

❖ Learning Outcome

- Understand basic terminology and fundamental concepts in research
- Comprehend basic flow of research process and formulate it for individual research
- Identify the characteristics of quantitative and qualitative research in the six steps
- Discuss important ethical issues in conducting research
- Recognize skills needed to design and conduct research

Outline

- Definition of Research and Its Importance
- Phases of Research/Research Process
- The Characteristics of Quantitative and Qualitative Research in each Research Process
- Important Ethical Issues in Conducting Research
- Features of a Good Research Study

❖ Definition of Research and Its Importance

➢ What is Research?

✓ A systematic investigation to **gain new knowledge** from **already existing facts**.

✓ A **scientific understanding of existing knowledge** and **deriving new knowledge** that may be applied for the betterment of mankind.

✓ A process of steps used to **collect and analyze information** to **increase our understanding of a topic or issue**.

❖ Definition of Research and Its Importance

- Research should always aim at providing **efficient solutions** to routine **problems**.
- Research should carefully choose the **appropriate research method** and follow a research process by referring to existing theories.
- Research **comprises** of defining and redefining *problems*, formulating the *hypothesis* for suggested solutions, collecting, organizing and evaluating *data*, making deductions and reaching conclusion and further testing the conclusion whether they fit into formulating the hypothesis.

❖ Definition of Research and Its Importance

- The **main purpose of research** is to find solutions to unsolved problems using scientific procedures and also to understand various phenomena scientifically.
- At a general level, research consists of three steps:



❖ Definition of Research and Its Importance

Example 1:

Research Question: Does watering plants daily help them grow taller?

Collect Data to Answer the Question:

- Before Watering Daily:
  - Plant Height Measurement: Average Height = 10cm
- After Watering Daily for One Week
  - Plant Height Measurement: Average Height = 15cm

❖ Definition of Research and Its Importance

Present an Answer to the Question:

- Findings:
  - Height Comparison: Before Watering Daily – After A Week
- Conclusion
  - Height Increase: Plant height increased by 5 cm after watering daily

❖ Definition of Research and Its Importance

Example 2:

Research Question: How can social media be used most effectively for marketing a new mobile app?

Collect Data:

This research can involve various data collection methods:

- **Literature Review**:
  - Conduct a thorough review of academic journals, research papers, and industry reports on social media marketing for mobile apps

❖ Definition of Research and Its Importance

- **Literature Review (Cont.)**:
  - Identify existing knowledge, theoretical frameworks, and successful strategies documented in prior research
  - Focus on keywords like “social media marketing,” “mobile apps,” “user acquisition,” and “engagement”
  - Utilize academic databases (ScienceDirect, JSTOR, Google Scholar) to find relevant publications
  - Critically analyze existing research, identify limitations, and explore potential gaps in knowledge

❖ Definition of Research and Its Importance

- **Surveys**:
  - Design surveys targeting mobile app developers and marketing professionals
  - Ask about their social media marketing strategies for new apps and what tactics yielded the best results (increased downloads, user engagement)
- **Case Studies**:
  - Analyze successful social media campaigns for new mobile apps across different platforms
  - Identify common themes, content types (videos, infographics, polls), and strategies used (platform targeting, influencer marketing)

❖ Definition of Research and Its Importance

- **Social Listening**:
  - Utilize social media listening tools to monitor online conversations about social media app marketing trends.
  - Track popular hashtags, influencer mentions, and audience interests related to app marketing.

❖ Definition of Research and Its Importance

Present an Answer to the Question:

Data Analysis:

- Analyze the data collected from all sources (literature review, surveys, case studies, social listening)
- Identify key factors that contribute to successful social media campaigns for new mobile apps. This might include:
  - ✓ **Platform Targeting**: Focusing on the social media platforms most frequented by your target audience
  - ✓ **Content Strategy**: Creating engaging content formats (videos, infographics, polls) that showcase the app's features and benefits
  - ✓ **Influencer Marketing**: Partnering with relevant influencers to reach a wider audience within your niche
  - ✓ **Social Media Advertising**: Utilizing targeted social media ads to drive app downloads and user acquisition.

❖ Definition of Research and Its Importance

Present an Answer to the Question:

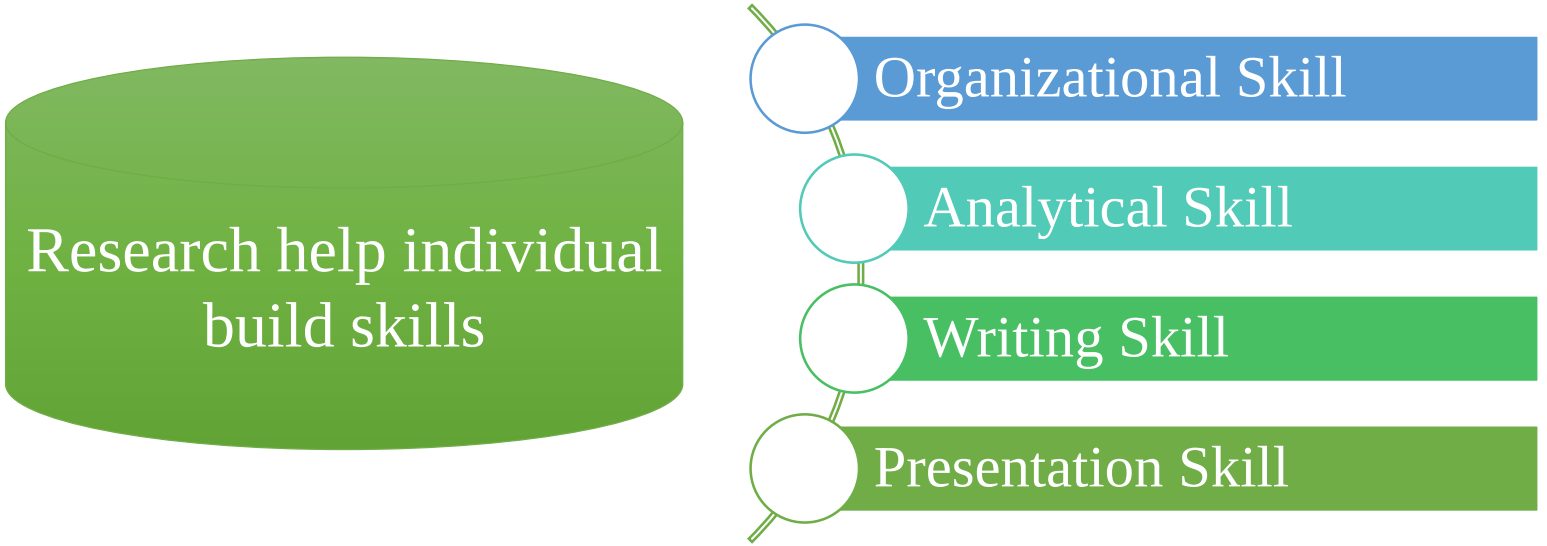
Presenting an Answer:

- Form a conclusion that addresses the research question

Example: Social media can be used most effectively for marketing a new mobile app by implementing a strategic approach that combines targeted platforms, engaging content, influencer partnerships, and data-driven advertising.

❖ Definition of Research and Its Importance

The importance of Research:



➢ Research can help to solve problem and corrects the misbelief, generate new concepts and broadens perspectives and perceptions

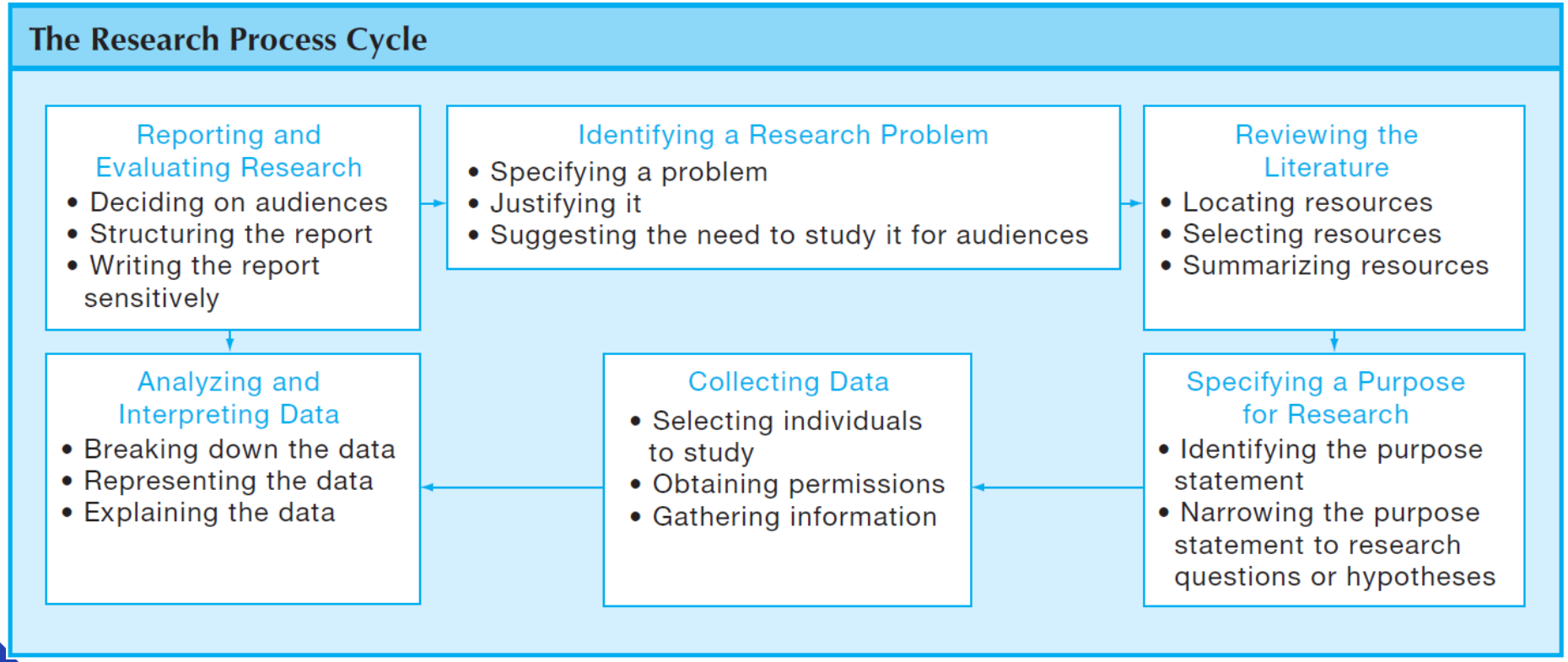
❖ Definition of Research and Its Importance

Ingredients for Research:



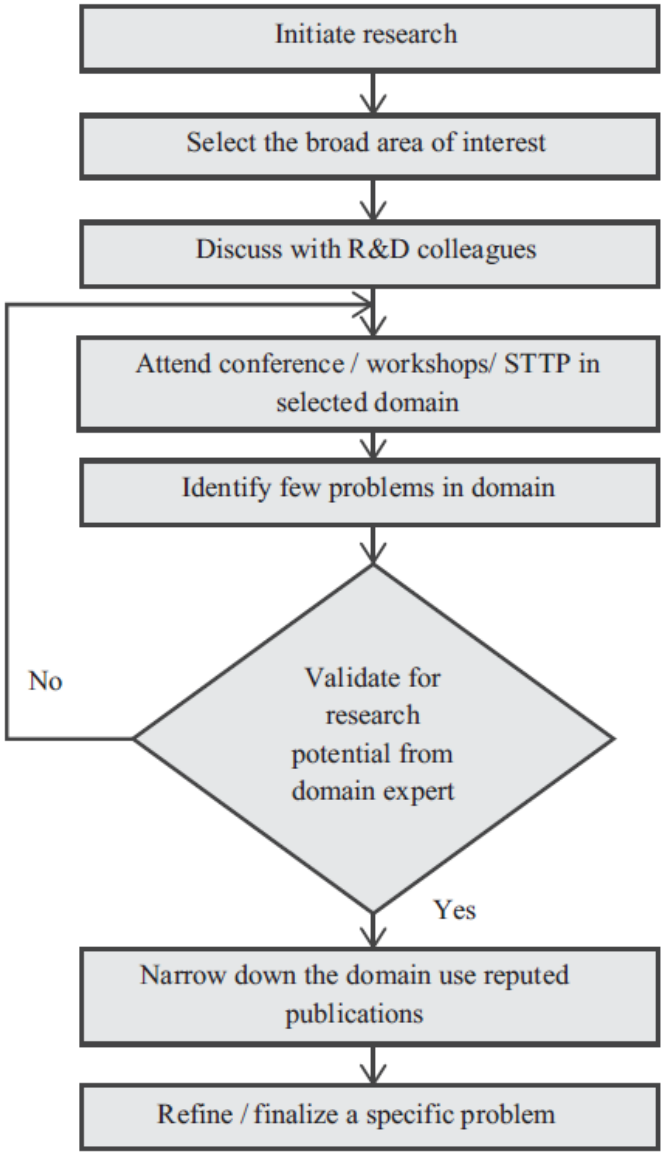
❖ Phases of Research/Research Process

➢ When researchers conduct a study, they proceed through a distinct set of steps. The process of research consists of **six** steps:



❖ Phases of Research/Research Process

➢ Before starting with research; researcher should be equipped with all the tools and domain knowledge required for research. The figure here indicates the general guideline for selection of research problem.



❖ Phases of Research/Research Process

➢ Researcher should prepare the research tentative plan of research work.

Month (M) / Tasks	M1 to M6	M7 to M12	M13 to M18	M19 to M24	M25 to M30	M31 to M36	M37 to M42
Selection of Domain							
Formulating a Research problem (tentative) & Identification of Keywords							
Literature Survey							
Redefining Research Problem & Objectives (Final)/ Formulating							
Hypothesis							
Preparing Research Proposal							
Identification of variable/parameters & Research Design							
Data Collection & representation Analysis							
Testing of proposed design on collected data/Hypothesis testing							
Results & Analysis							
Research Report/Theses writing/Publication/Patent							

❖ The Characteristics of Quantitative and Qualitative Research in each Research Process

➢ In **quantitative research**, the major characteristics are the following:

- Describing a research problem through a description of trends or a need for an explanation of the relationship among variables
- Providing a major role for the literature through suggesting the research questions to be asked and justifying the research problem and creating a need for the direction (purpose statement and research questions or hypotheses) of the study

The Characteristics of Quantitative and Qualitative Research in each Research Process

- Creating purpose statements, research questions, and hypotheses that are specific, narrow, measurable, and observable
- Collecting numeric data from a large number of people using instruments with preset questions and responses
- Analyzing trends, comparing groups, or relating variables using statistical analysis and interpreting results by comparing them with prior predictions and past research
- Writing the research report using standard, fixed structures and evaluation criteria and taking an objective, unbiased approach

❖ The Characteristics of Quantitative and Qualitative Research in each Research Process

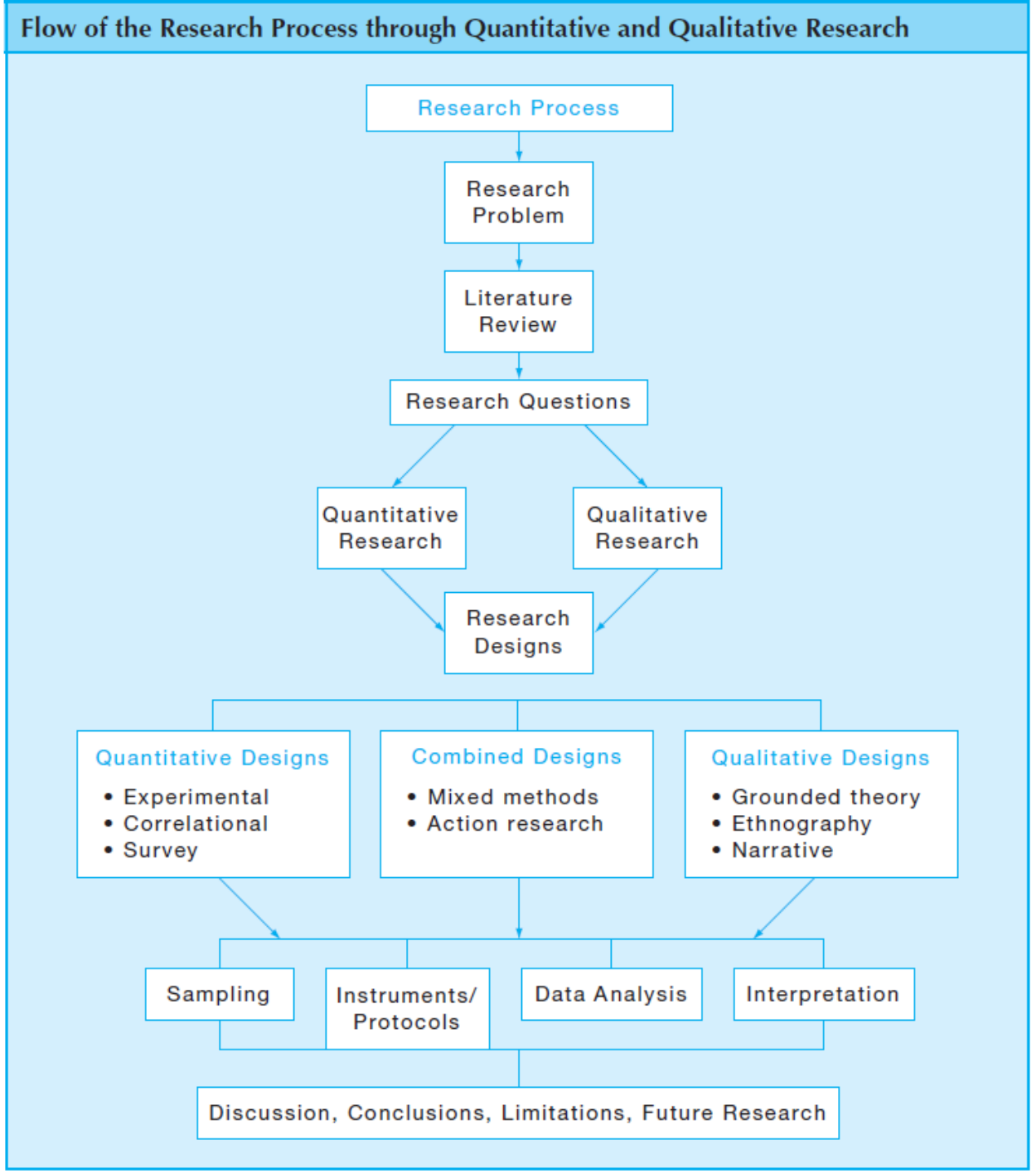
➢ In **qualitative research**, we see different major characteristics at each stage of the research process:

- Exploring a problem and developing a detailed understanding of a central phenomenon
- Having the literature review play a minor role but justify the problem
- Stating the purpose and research questions in an open-ended way to capture the participants' experiences

The Characteristics of Quantitative and Qualitative Research in each Research Process

- Collecting data that is based on words (e.g., from interviews) or images (e.g., photographs) from a small number of individuals so that the participants' views are obtained
- Analyzing the data for description and themes using text analysis and interpreting the larger meaning of the findings
- Writing the report using flexible, emerging structures and evaluative criteria and including the researchers' subjective reflexivity and bias

- **The Characteristics of Quantitative and Qualitative Research in each Research Process**



❖ Important Ethical Issues in Conducting Research

- Research ethics focus on the moral principles that researchers must follow in their respective fields of research



Image Source: Ethics

❖ Important Ethical Issues in Conducting Research

- Learn about the procedures involved in applying for approval from your campus institutional review board
- Recognize guidelines from professional associations
- Use ethical practices throughout research
- Use respectful data collection procedures
- Show respect to audiences who read and use research study information

Image Source: Ethics

❖ Features of a Good Research Study

➢ Some of the important features of good research are:

- Research purpose should be clearly defined
- Procedure for the research should be detailed sufficiently which should help the other to continue the work by referencing our work
- Research work should be carefully planned to get the results in related to the specified objectives
- Reports should be created by a researcher stating that what was the procedure adopted for completing the work which should also include errors in their findings
- Conclusions should be confident to those justified by the data of research

Q&A